Table 61. Minimum I2CCLK frequency

Symbol	Parameter	Condition		Typ	Unit
$f_{I2CCLK(min)}$	Minimum I2CCLK frequency for correct operation of I2C peripheral	Standard-mode		2	MHz
		Fast-mode	Analog filter enabled DNF = 0	9	
			Analog filter disabled DNF = 1	9	
		Fast-mode Plus	Analog filter enabled DNF = 0	19	
			Analog filter disabled DNF = 1	16	

The SDA and SCL I/O requirements are met with the following restrictions: the SDA and SCL I/O pins are not “true” open-drain. When configured as open-drain, the PMOS connected between the I/O pin and V_{DDIO1} is disabled, but is still present. Only FT_f I/O pins support Fm+ low-level output current maximum requirement. Refer to *Section 5.3.13: I/O port characteristics* for the I2C I/Os characteristics.

All I2C SDA and SCL I/Os embed an analog filter. Refer to the following table for its characteristics:

Table 62. I2C analog filter characteristics

Symbol	Parameter(1)	Min	Max	Unit
t_{AF}	Maximum pulse width of spikes that are suppressed by the analog filter	50(2)	150(3)	ns

1. Evaluated by characterization. Not tested in production.
2. Spikes with widths below t_{AF} (min) are filtered.
3. Spikes with widths above t_{AF} (max) are not filtered.

USART (SPI mode) characteristics

Unless otherwise specified, the parameters given in *Table 63* for USART are derived from tests performed under the ambient temperature, f_{PCLKx} frequency and supply voltage conditions summarized in *Table 24: General operating conditions*. The additional general conditions are:

- OSPEEDRy[1:0] set to 10 (output speed)
- capacitive load C = 30 pF
- measurement points at CMOS levels: 0.5 × V_{DD}

Refer to *Section 5.3.13: I/O port characteristics* for more details on the input/output alternate function characteristics (NSS, CK, TX, and RX for USART).